Maths IA-1 — Master Worksheet

All 6 chapters · all levels · worked answers included

Generated 25 Oct 2026 · exam pattern reference: 35 marks / 90 min written (assumed from Contest-1)

How to use: attempt in order. Each chapter: Basic (bank the easy marks) → Exam level → Hard → Twist (the central-team style). Do the **Mini Mock** under 40-min timer with no notes. Answers with full worked solutions are at the end — do NOT peek before attempting.

Chapter 1 — Algebraic Expressions

Coverage: expansion · factorisation · identities · substitutions · exponent rules

1. Simplify: $3x + 5 - x + 2$
2. Factorise: $x^2 + 7x + 12$
3. Expand: $(x + 2)(x - 5)$
4. Simplify: $(a + b)^2 - (a - b)^2$
5. If $x + 1/x = 5$, find $x^2 + 1/x^2$
6. Factorise fully: $x^4 - 16$
7. If $a + b = 7$ and $ab = 10$, find $a^2 + b^2$ and $a^3 + b^3$
8. Simplify: $(x^2y^3)^4 \div (x^3y^5)$
9. Find the area left when four 1×1 squares are cut from the corners of a square of side x
10. **(Twist)** Prove that $(2n + 1)^2 - (2n - 1)^2$ is always divisible by 8 for any integer n

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Chapter 2 — Quadratic Equations

Coverage: factorisation · completing the square · discriminant · roots (sum/product) · word problems

1. Solve: $x^2 + 7x + 12 = 0$
2. Solve: $2x^2 - 5x - 3 = 0$
3. Find the discriminant of $3x^2 - 2x + 5 = 0$ and state the root nature
4. For what value of k does $kx^2 + 4x + 1 = 0$ have equal roots?
5. If α, β are roots of $x^2 - 6x + 8 = 0$, find $\alpha^2 + \beta^2$ and $1/\alpha + 1/\beta$
6. Write $x^2 + 6x + 2$ in completed-square form
7. Two numbers sum to 12 and their product is 32. Find them
8. Solve: $x^2 + 1/x^2 = 79$ (hint: find $x + 1/x$ first)
9. **(Twist)** For which values of a does $ax^2 - 2x + 3 = 0$ have BOTH roots positive?
10. **(Twist)** If the equation $x^2 + bx + 16 = 0$ has equal roots, find b. What is the root?

Chapter 3 — Inequalities

Coverage: linear · absolute value · quadratic sign · rational · "for all values" logic

1. Solve: $2x - 3 > 7$
2. Solve: $-3x + 4 \leq 10$ (watch the flip)
3. Solve: $x^2 - 5x + 6 < 0$
4. Solve: $|x - 3| \leq 5$
5. Solve: $(x - 1)/(x + 2) \geq 0$
6. Solve: $5 - 2x \geq x + 2$
7. **(Twist)** Find all real k such that $x^2 + 2x + k > 0$ for ALL real x
8. **(Twist)** Solve: $-5 < 1/x < 2$
9. **(Twist)** Find the SUM of all integers satisfying $x^2 - 9 < 0$
10. **(Hard)** Solve: $(x - 3)/(x + 1) < 2$

Chapter 4 — Logarithms & Exponents

Coverage: definition · laws · change of base · equations · word/application

1. Evaluate: $\log_2 32$
2. Evaluate: log 1000 (base 10)
3. Evaluate: $\log_2 6 - \log_2 3$
4. Express log 45 in terms of log 3 and log 5
5. Evaluate: $\log_9 27$
6. Solve: $\log_2(x + 1) = 3$
7. Solve: $\log x + \log(x - 3) = 1$
8. Evaluate: $\log_2 3 \times \log_3 8$
9. If $2^x = 7$, write x in log form
10. **(Twist)** How many digits does 5^{20} have? (given $\log_{10} 5 = 0.6990$)

Chapter 5 — Divisibility, GCD & LCM

Coverage: Euclidean algorithm · gcd×lcm · counting multiples · remainder patterns · word problems

1. Find gcd(48, 18)
2. Find lcm(12, 18)
3. If gcd(a, b) = 8 and lcm(a, b) = 96 and a = 24, find b
4. Use the Euclidean algorithm to find gcd(1071, 462)
5. Find the last two digits of 2^{100}
6. Buses to the same stop leave every 15 min and every 20 min (both at 8:00). At what time do they next leave together?
7. Find the largest number that divides 240, 320 and 400 exactly

8. How many numbers from 1 to 100 are divisible by 6 OR by 8?

9. Is $2^{21} - 1$ divisible by 7? Justify in one line.

10. **(Twist)** $n = 5^2 \cdot 7^3 \cdot 11$. How many positive divisors does n have?

Chapter 6 — Congruence, Modular Arithmetic & Euclidean Algorithm

Coverage: mod operation · cyclicity · big powers mod n · inverses · linear congruences · divisibility proofs

1. Find 100 mod 7

2. Find the last digit of 3^{250}

3. Solve: $x + 2 \equiv 0 \pmod 7$ — give the general solution

4. Find 2^{100} mod 7

5. Find 7^{222} mod 5

6. Find the modular inverse of 3 modulo 11

7. Solve: $3x \equiv 6 \pmod 9$

8. Prove: $n^3 - n$ is always divisible by 3

9. Find the last two digits of 3^{2024}

10. **(Twist)** Find the remainder when 7^{99} is divided by 100. (HARD — use $\lambda(100)=20$ idea or repeated squaring; if stuck, do mod 4 and mod 25 separately and combine.)

MINI MOCK — exam conditions, 40 minutes, no notes

(Mixed, roughly half the paper's coverage)

A. MCQ (2 marks each):

1. gcd(18, 45) = ? (a) 3 (b) 6 (c) 9 (d) 15

2. If $x + 1/x = 3$ then $x^2 + 1/x^2$ = ? (a) 5 (b) 7 (c) 9 (d) 11

3. $\log_4 64$ = ? (a) 2 (b) 3 (c) 4 (d) 6

4. Number of positive divisors of 60 = ? (a) 6 (b) 8 (c) 10 (d) 12

B. Short (3 marks each):

5. Solve: $2x^2 - 7x + 3 = 0$

6. Solve: $3x - 4 < 5x + 2$

7. Find remainder of $4^{50} \div 7$

C. Working (write full reasoning — 5 marks each):

8. α, β are roots of $2x^2 - 5x - 3 = 0$. Find $\alpha^3 + \beta^3$.

9. Find the greatest number that divides 64, 127 and 187 leaving the same remainder in each case.

10. Prove that for any integer n, $n^4 - n^2$ is divisible by 12.

ANSWERS — full worked solutions

Chapter 1

1. **$2x + 7$**
2. **$(x + 3)(x + 4)$** — need product 12, sum 7 → 3, 4
3. **$x^2 - 3x - 10$**
4. **$4ab$** — $(a^2+2ab+b^2) - (a^2-2ab+b^2) = 4ab$
5. **23** — $x^2 + 1/x^2 = (x + 1/x)^2 - 2 = 25 - 2 = 23$
6. **$(x - 2)(x + 2)(x^2 + 4)$** — difference of squares twice
7. **$a^2+b^2 = 29$; $a^3+b^3 = 133$** — $a^2+b^2 = 49 - 20 = 29$; $a^3+b^3 = (a+b)^3 - 3ab(a+b) = 343 - 210 = 133$
8. **x^5y^7** — $(x^8y^{12}) \div (x^3y^5)$
9. **$x^2 - 4$** — original x^2, minus 4 ones
10. $(2n+1)^2 - (2n-1)^2 = [(2n+1)+(2n-1)][(2n+1)-(2n-1)] = (4n)(2) = 8n \rightarrow$ divisible by 8 ∎

Chapter 2

1. **$x = -3$ or -4**
2. **$x = 3$ or $-1/2$** — $(2x + 1)(x - 3) = 0$
3. **$D = 4 - 60 = -56 < 0 \rightarrow$ no real roots**
4. **$k = 4$** — $D = 16 - 4k = 0$
5. **$\alpha^2+\beta^2 = 20$; $1/\alpha+1/\beta = 3/4$** — sum 6, product 8; $\alpha^2+\beta^2 = 36 - 16$; $1/\alpha+1/\beta = 6/8$
6. **$(x + 3)^2 - 7$**
7. **4 and 8** — $t^2 - 12t + 32 = 0 \rightarrow (t-4)(t-8)$
8. **$x + 1/x = 9$** (taking positive root since $x^2+1/x^2 = 79 > 0 \rightarrow (x+1/x)^2 = 81$). *If asked x: $x = (9 \pm \sqrt{77})/2$*
9. **$0 < a \leq 1/3$** — sum $= 2/a > 0 \rightarrow a > 0$; product $= 3/a > 0$ (holds); $D \geq 0 \rightarrow 4 - 12a \geq 0 \rightarrow a \leq 1/3$
10. **$b = \pm 8$, root $= \mp 4$** — $D = b^2 - 64 = 0 \rightarrow b = \pm 8$; if $b = 8$: $x = -4$; if $b = -8$: $x = 4$

Chapter 3

1. **$x > 5$**
2. **$x \geq -2$** (flip when dividing by -3)
3. **$2 < x < 3$** — parabola negative between roots
4. **$-2 \leq x \leq 8$**
5. **$x \geq 1$ or $x < -2$** ($x = -2$ excluded — denominator)
6. **$x \leq 1$**
7. **$k > 1$** — $D = 4 - 4k < 0 \rightarrow k > 1$ (strict: if $k = 1$, zero at $x = -1$, fails)
8. **$x > 1/2$ or $x < -1/5$** — solve both halves: $1/x < 2 \rightarrow x > 1/2$ or $x < 0$; $1/x > -5 \rightarrow x > 0$ or $x < -1/5$; intersect
9. **0** — integers $-2, -1, 0, 1, 2 \rightarrow$ sum $= 0$
10. $(x - 3) < 2(x + 1)$ if $x + 1 > 0$: $x - 3 < 2x + 2 \rightarrow x > -5$; with $x > -1 \rightarrow x > -1$. If $x + 1 < 0$: inequality flips: $x - 3 > 2x + 2 \rightarrow x < -5$. **Solution: $x > -1$ or $x < -5$** (check $x = -2$: $(-5)/(-1) = 5 < 2$? NO → excluded, consistent)

Chapter 4

1. **5** ($2^5 = 32$)
2. **3**
3. **1** ($\log_2 2 = 1$)
4. **2 log 3 + log 5** ($45 = 9 \times 5$)
5. **3/2** (9^{3/2} = 27)
6. **x = 7**
7. **x = 5** (reject x = –2 — log of negative)
8. **3** (chain: $\log_2 3 \times \log_3 8 = \log_2 8 = 3$)
9. **x = $\log_2 7$** (equivalently log 7 / log 2)
10. **14 digits** — digits = $\lfloor 20 \times 0.6990 \rfloor + 1 = \lfloor 13.98 \rfloor + 1 = 14$

Chapter 5

1. **6** ($48 = 2 \cdot 18 + 12$; $18 = 1 \cdot 12 + 6$; $12 = 2 \cdot 6$)
2. **36**
3. **b = 32** — gcd × lcm = 192 = 24 × b
4. **21** — $1071 = 2 \times 462 + 147$; $462 = 3 \times 147 + 21$; $147 = 7 \times 21 + 0$
5. **76** — powers of 2 mod 100 cycle every 20 from 2^2; $2^{100} \rightarrow$ position (98 mod 20) = 18 $\rightarrow 2^{20}$ = ...76
6. **9:00 AM** — lcm(15, 20) = 60 min
7. **80** — gcd(240, 320, 400)
8. **24** — 16 + 12 – 4 (6s: 16, 8s: 12, 24s: 4)
9. **Yes** — $2^3 \equiv 1 \pmod 7 \rightarrow 2^{21} = (2^3)^7 \equiv 1 \pmod 7 \rightarrow 2^{21} - 1 \equiv 0$ ∎
10. **24 divisors** — (2+1)(3+1)(1+1) = 24

Chapter 6

1. **2** ($100 = 14 \times 7 + 2$)
2. **9** — cycle of 3's last digit is 4; 250 mod 4 = 2 $\rightarrow 3^2 = 9$
3. **x ≡ 5 (mod 7)** → x = 5, 12, 19, ...
4. **2** — $2^3 \equiv 1 \pmod 7$; $100 = 33 \times 3 + 1 \rightarrow 2^{100} \equiv 2^1 \equiv 2$
5. **4** — $7 \equiv 2 \pmod 5$; cycle of 2 mod 5 is 4; 222 mod 4 = 2 $\rightarrow 2^2 = 4$
6. **4** — $3 \times 4 = 12 \equiv 1 \pmod{11}$
7. **x ≡ 2 (mod 3)** → x = 2, 5, 8 (gcd 3 divides 6 → 3 solutions mod 9)
8. $n^3 - n = (n - 1)n(n + 1)$ = three consecutive integers; one is a multiple of 3 → divisible by 3 ∎
9. **81** — order of 3 mod 100 is 20; 2024 mod 20 = 4 $\rightarrow 3^4 = 81$
10. **43** — two pieces: (i) mod 4: $7 \equiv -1 \rightarrow 7^{99} \equiv -1 \equiv 3 \pmod 4$. (ii) mod 25: $7^2 = 49 \equiv -1 \rightarrow 7^4 \equiv 1 \pmod{25} \rightarrow 7^{99} =$ 7^(4×24)·$7^3 \equiv 7^3 = 343 \equiv 18 \pmod{25}$. Combine: $x \equiv 3 \pmod 4$, $x \equiv 18 \pmod{25} \rightarrow x = 18 + 25k$, 18 mod 4 = 2 → 2 + $k \equiv 3 \rightarrow k = 1 \rightarrow x = 18 + 25 =$ **43**

MINI MOCK

1. **(a) 3**

2. **(b) 7** (9 – 2)

3. **(b) 3**

4. **(d) 12** — $60 = 2^2 \cdot 3 \cdot 5 \rightarrow (2+1)(1+1)(1+1) = 12$

5. x = 3 or 1/2 — $(2x - 1)(x - 3) = 0$

6. **x > –3** — $3x - 4 < 5x + 2 \rightarrow -6 < 2x \rightarrow x > -3$

7. **2** — $4^3 = 64 \equiv 1 \pmod 7$; $4^{50} = (4^3)^{16} \cdot 4^2 \equiv 1 \cdot 16 \equiv 2$

8. Roots: 3 and $-1/2 \rightarrow \alpha^3+\beta^3 = 27 - 1/8 =$ **215/8**. (Check: sum 5/2, product $-3/2 \rightarrow \alpha^3+\beta^3 = (5/2)^3 - 3(-3/2)(5/2) = 125/8 + 45/4 = 125/8 + 90/8 = 215/8$ ✓)

9. Method: N must divide ALL pairwise differences → 127−64 = 63, 187−127 = 60, 187−64 = 123. gcd(63, 60, 123) = **3**. Check remainders: 64 mod 3 = 1, 127 mod 3 = 1, 187 mod 3 = 1 — equal ✓. **Answer: 3**

10. $n^4 - n^2 = n^2(n - 1)(n + 1)$. **By 3:** the three consecutive integers (n−1), n, (n+1) contain a multiple of 3, and $n^2(n-1)(n+1)$ contains all three → divisible by 3. **By 4:** if n is even, n^2 is divisible by 4; if n is odd, both (n−1) and (n+1) are even → divisible by 4 (even by 8). Since divisible by 3 and 4, and gcd(3,4) = 1 → divisible by 12 ∎

Worksheet ends. Anything you got wrong → re-solve it WITHOUT looking at the solution, then find a cousin problem tonight.